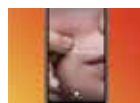




USB driver vulnerabilities

Researchers have used a new tool to identify vulnerabilities in the USB drivers of OSs. <https://dgit.in/jun20-90>



Instagram to share revenues

Instagram will be sharing revenues with creators through ads on the IGTV platform. <https://dgit.in/jun20-91>

bre optics work. It was way ahead of its time, because the entire backbone of the internet is essentially a huge network of photophones, transmitting data instead of voice as Bell did. However, although we took his wireless communication device and turned it into a fibre (essentially wired) device, it did stoke people's imaginations and proved that wireless communication was possible.

Remember, Bell's photophone was a good 14 years before Guglielmo Marconi started working on the radio. Marconi would eventually be awarded a patent for it in 1896.

Fast forward to 1973, and we've all heard of the story of Martin Cooper, a researcher at Motorola, placing a call from the first ever handheld mobile device to his rival, Joel Engel of Bell Labs. If you haven't heard the story, look it up, it's pretty funny. The first generation of cellular networks (1G) took off just after 1981 across the world. That's a mere 40 years ago! However, that was all voice telephony.

It was with 3G, in the early 2000s that mobile internet usage really took off, and slowly kickstarted the mobile internet trends that we're still surfing. When this magazine started in 2001, 3G was the hottest thing we'd heard of, and Japan and NTT DoCoMo were words we used with awe. Today, with 4G everywhere, and 5G mobile networks growing rapidly in some parts of the world, a 3G connection would feel so slow we'd be calling our service providers to complain!

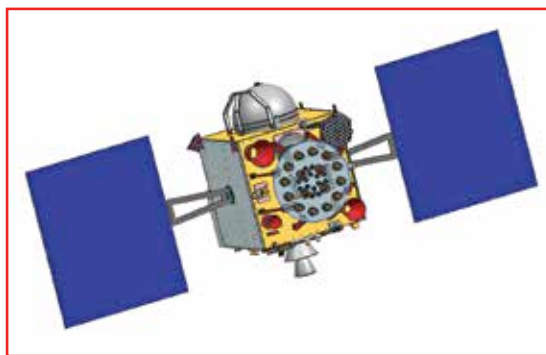
GPS

The global positioning system that we take for granted now, was started in the 1970s by the US military. Later, it was made available for the rest of the world, for free. Currently, the US, Russia, the EU, India, China and Japan have their own satellite navigation systems.

After the year 2000, when civilians were allowed to receive non-degraded signals from the US satellites (they had degraded signals as a security measure, to safeguard military installations), Sat-Nav / GPS really took off. Right until 2000 there was still merit in learning how to use a map if you drove across

countries. Technology changed that. Today, you have more accurate mapping on your phone than the military had in the early 1980s! And boy do we use it... to find our way everywhere, to tell others where we are, to call ride-shares to our location, etc.

Pretty much every app on your phone wants location data because everything is tailored to your physical location, and this is only going to increase as things get more local and everything is targeted at you more precisely.



A render of a typical IRNSS satellite that enables our NavIC satellite navigation system

BIG BROTHER

Targeting is our cue to talk about big brother-ish technology that's become the new normal for all of us. We used to think it would be evil governments who were spying on citizenry, and while that's happening in many places around the globe as well, it's much more corporations with their products and services who are trying to gather data about you.

Why? Because the more that technology can help find out about you, the more you can be influenced into thinking / buying / voting a certain way. To some this doesn't seem like a big deal... if you're the type to want to buy a 75-inch QHD TV that costs Rs 3 Lakh, then is there anything wrong with a company that makes such a TV from showing you ads about their TV? Probably not. However, if you were talking to your partner casually, and now a machine somewhere is putting together a large list of keywords that describe the conversation... then selling that data to a bunch of companies, based on what you spoke about. And not to everyone,

but to the highest bidder... that's when you see what's wrong with the picture.

Essentially, with control belonging to a small group of companies, technology is essentially a monopoly now, and the companies who control everything are happily surrounding you, walling you off from all of their competition. This is where we find ourselves, and honestly there is no clear way out of this for all of us. The best thing we can do is to ensure that we don't allow monopolies to thrive.

No one is going to give up their Facebook/Instagram account, but maybe you should try other social platforms as well, just to keep them pitted against one another. Either way, it certainly is a massive achievement by these companies, to have walled billions of us into their services, and created a totally new normal for society at large...

LHC

Instead of focussing on that last negative, it's time to jump back to something a lot more positive. The fact that something like the LHC could exist or be built, with multiple countries from across the globe participating, some as providers of money, or technology, land etc, and others sending their scientists and expertise, and even more taking data and doing analytical work, all towards a common goal of understanding our universe better. So many countries putting science ahead of politics gives us hope, and to be honest, the reaction worldwide to that-virus-that-shan't-be-named-because-we're-sick-of-hearing-about-it has brought a similar level of hope of collaboration and value for science.

As we've seen from history, it's scientific breakthroughs like the ones that happen at the LHC that allow us to push the limits of what technology can do. And the LHC discovered the Higgs Boson, which is how everyone knows the LHC. What most people don't know